

EGA-102 Group Design Project

Design Report

Production of Phenol via Chlorination and Hydrolysis of Benzene

Authors	Jibran Ahmed (2449068) Mustafa Qureshi (2543631) Noel Renni (2517447) Mustafa Abdelkarim (2512713)
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1. Summary

Principal Author: Jibran Ahmed (2449068), Noel Renni (2517447), Mustafa Qureshi (2543631), Mustafa Abdelkarim (2512713)

This report presents the group design of a chemical process to produce phenol via the chlorination of benzene to monochlorobenzene (MCB) followed by the MCB hydrolysis to phenol. The design is based on a process basis of 100 kmol/h of benzene in the chlorinator feed and was carried out under steady-state conditions at 1 bar.

The overall material balance confirms that the process achieves approximately 97.2% overall conversion of fresh benzene feed, with the remaining benzene leaving as a minor impurity in the K2 column. We utilised two parallel chlorination reactors operating isothermally at 270°C which convert benzene to MCB at 40% conversion per pass, with 8% of the converted benzene combusting to CO₂ and H₂O. The combined reactor effluent is cooled via a waste-heat boiler and a secondary heat exchanger, producing 2,052 kg/h of dry saturated steam at 100°C. The process liquid undergoes phase separation in a decanter, liquid-liquid extraction, and sequential distillation in two columns (K1 and K2). K1 recovers phenol at 99 mol% purity, and K2 separates benzene (97 mol% top product) from MCB (97 mol% bottom product) for recycling and further processing.

The hydrolysis reactor operates at 310°C with 30% conversion per pass and includes an MCB recycle loop. Energy balance calculations indicate that both reactors are exothermic and require heat removal: the chlorination reactor requires a cooling duty of approximately 4,735 kJ/s, while the hydrolysis reactor requires a cooling duty of approximately 147 kJ/s. The HCl heat exchanger duty is 17.2 kW and the boiler duty is 1,287 kJ/s.

The vapour-liquid equilibrium (VLE) analysis of the benzene-MCB distillation column K2 using Raoult's Law and Antoine data, determined that 9 theoretical stages (8 trays plus a reboiler) is sufficient at a reflux ratio of 0.489 (1.5 times the minimum of 0.326) for the actual process feed composition (60.8 mol% benzene). Individual analyses for feed compositions of 20, 30, 40, and 50 mol% MCB, representing scenarios of improved reactor conversion, show that the required reflux ratio increases significantly with MCB content, ranging from 0.165 to 0.726 with the fixed stage count of 9, highlighting the importance of flexible column operation.

Key safety considerations include the flammability and carcinogenicity of benzene, the toxicity and corrosivity of HCl, and the high operating temperatures throughout the plant. Appropriate measures such as containment, monitoring, and personal protective equipment are essential for safe operation.

2. Design Overview

Principal Author: Jibran Ahmed (2449068), Mustafa Qureshi (2543631)

2.1 Process Description

The process is designed to produce phenol from benzene using a two-stage reaction scheme. In the first stage, benzene is chlorinated using hydrochloric acid and oxygen (supplied as air) to produce monochlorobenzene (MCB). In the second stage, MCB is hydrolysed with steam to produce phenol and regenerate HCl, which is recycled back to the chlorination reactors.

2.1.1 Chlorination Section

Fresh benzene and recycled benzene-rich streams from distillation column K2 are combined with air and hydrochloric acid (both fresh and recycled) to form the reactor feed (100 kmol/h total benzene, with an air-to-benzene molar ratio of 3:1 and HCl which aligns with the stoichiometric requirements). Both chlorination reactors (R1 and R2) operate isothermally at 270°C. R1 produces half the mass flow of R2, though both share the same molar product composition. The chlorination reaction (Equation 1) and the combustion side-reaction (Equation 2) proceed simultaneously, with 40% of benzene converting per pass and 8% of converted benzene combusting.

Equation 1 (Chlorination): $\text{C}_6\text{H}_6 + \frac{1}{2}\text{O}_2 + \text{HCl} \rightarrow \text{C}_6\text{H}_5\text{Cl} + \text{H}_2\text{O}$

Equation 2 (Combustion): $\text{C}_6\text{H}_6 + \frac{15}{2}\text{O}_2 \rightarrow 6\text{CO}_2 + 3\text{H}_2\text{O}$

2.1.2 Heat Recovery and Cooling

The hot combined reactor effluent (at 270°C) passes to a waste-heat boiler where it is cooled to 150°C, generating dry saturated steam at 100°C. This steam is used in a heat exchanger (HEX-1) to preheat the recycled HCl stream before it enters the chlorination reactors. The partially cooled reactor products then pass through a second heat exchanger (HE-2) where they are further cooled to 65°C. At this temperature, non-condensable gases (N_2 , O_2 , CO_2) are removed through a vent, and the remaining liquid is fed to a decanter.

2.1.3 Separation and Extraction

The decanter separates the process stream into an organic phase (benzene and MCB) and an aqueous phase (water and dissolved HCl). The aqueous phase is sent to a holding vessel; an overflow system controls the water level, with excess discharge to drain (limited to 550 kg/h maximum). The organic phase enters an extractor, where phenol is recovered from the hydrochloric acid stream arriving at 50°C from the hydrolysis unit. The combined organic stream from the extractor, containing benzene, MCB, and phenol, is fed to distillation column K1 at 89°C.

2.1.4 Distillation Section

Distillation column K1 separates phenol (as the bottoms product at 99 mol% purity) from the lighter benzene and MCB mixture (as the top product), operating at a reflux ratio of 2. The K1 top product, a binary benzene-MCB mixture (60.8 mol% benzene, 39.2 mol% MCB on a mole basis), is fed to K2, which produces a 97 mol% benzene distillate (recycled to the reactors) and a 97 mol% MCB bottom stream (fed to the hydrolysis reactor).

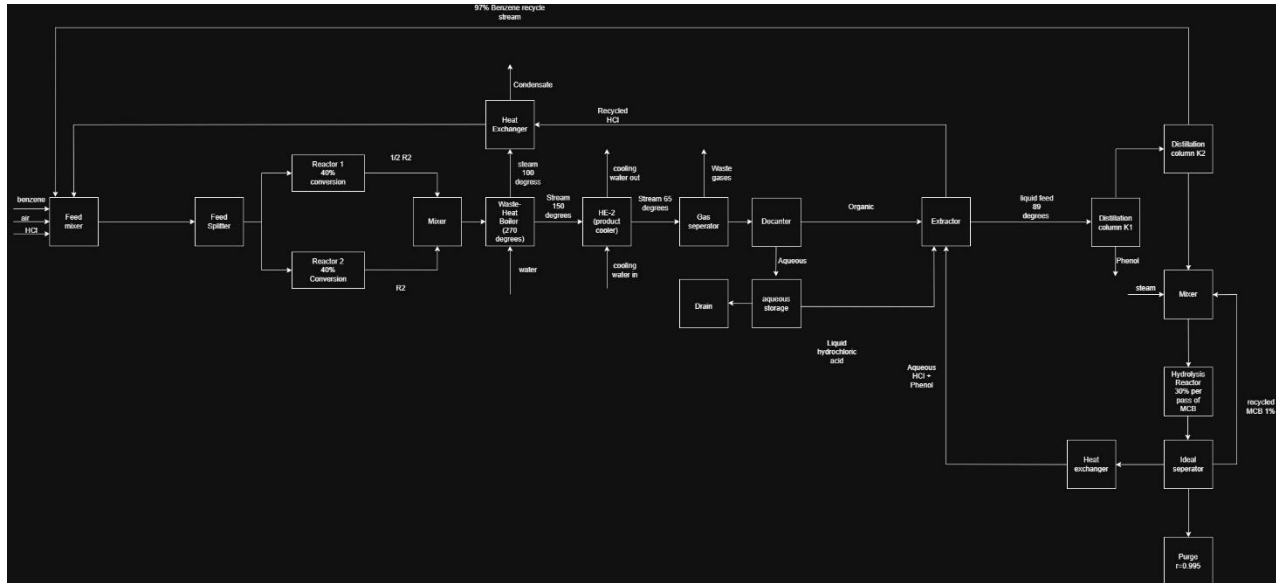
2.1.5 Hydrolysis Section

The MCB-rich bottoms from K2 is fed to the hydrolysis reactor (R3) at 310°C, together with steam at a molar steam-to-MCB feed ratio of 3:2. The reaction achieves 30% conversion per pass:

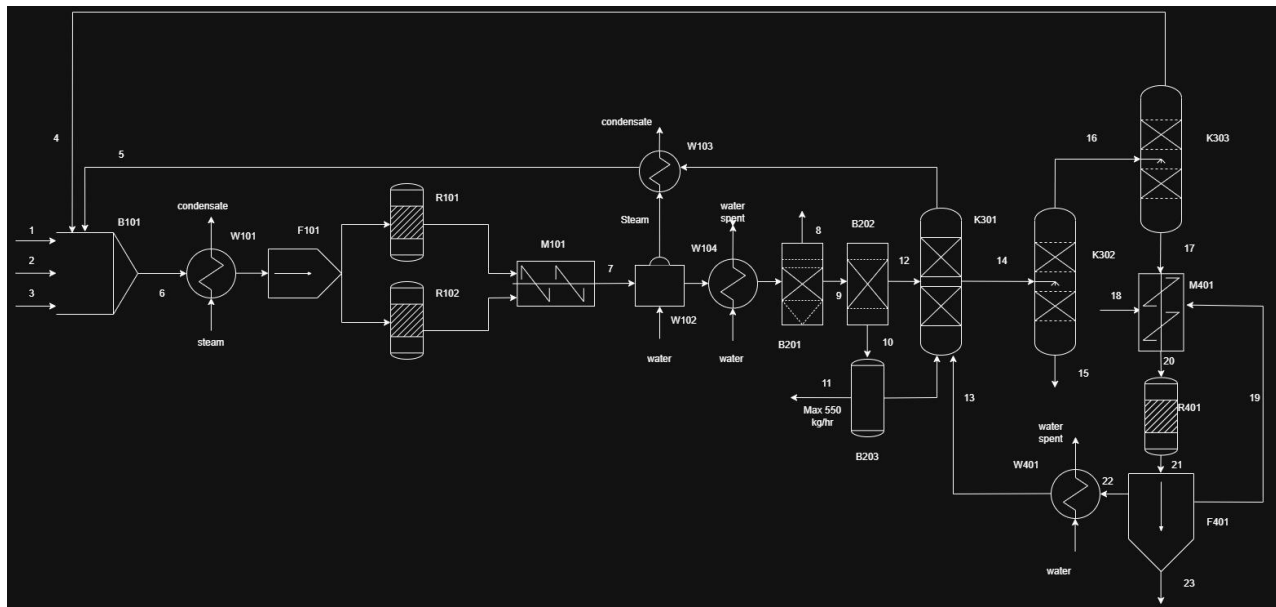
Equation 3 (Hydrolysis): $C_6H_5Cl + H_2O \rightarrow C_6H_5OH + HCl$

An ideal separator downstream recovers unconverted MCB at 99.9% purity for recycle back to R3, with a small purge ($r = 0.995$) to prevent volatile build-up. The phenol product and regenerated HCl leave the separator for further processing and recycling respectively.

2.2 Block Flow Diagram



2.3 Process Flow Diagram



3. Material Balance

Principal Author: Mustafa Qureshi (2543631), Noel Renni (2517447)

The overall material balance was performed on a basis of 100 kmol/h of benzene in the chlorinator mixed feed, operating under steady-state conditions at 1 bar. The following assumptions were applied:

- The process operates continuously at steady state, so all tank levels are constant.
- No vapour enrichment occurs at the vent, so no volatiles escape through the vent.
- Air composition is 79 mol% N₂ and 21 mol% O₂.
- The HCl aqueous feed is 25 wt%, with the remainder being water.
- No HCl gas leaves the chlorination unit.
- Hydrogen gas is not produced in the process.
- The ideal separator in the hydrolysis loop achieves complete recovery of MCB at 99.9% purity.
- The overall conversion of benzene within the process is approximately 97.2%, with the small remainder exiting as a bottoms impurity in the K2 distillation column.

3.1 Key Material Balance Results

Table 3.1 summarises the molar flow rates of key process streams. Stream numbers correspond to those on the Process Flow Diagram.

Table 3.1: Summary of Key Process Stream Molar Flow Rates (kmol/h), basis: 100 kmol/h benzene in chlorinator mixed feed

Stream	Benzene	MCB	HCl	H ₂ O	O ₂	N ₂	Phenol	Total
6 - Reactor Feed	100.0	1.82	36.80	21.40	63.0	237.0	-	460.0
7 - Reactor Output	60.0	38.62	-	67.80	20.6	237.0	-	443.2
9 - After Vent	60.0	38.62	-	67.80	-	-	-	166.4
10 - Aqueous (Dec.)	-	-	-	67.80	-	-	-	67.8
12 - Organic (Dec.)	60.0	38.62	-	-	-	-	-	98.6
16 - K2 Feed	60.0	38.62	-	-	-	-	-	98.6
4 - K2 Distillate (top)	58.86	1.82	-	-	-	-	-	60.7
17 - K2 Bottoms (bot.)	1.14	36.80	-	-	-	-	-	37.9
15 - Phenol product	-	-	-	-	-	-	36.38	36.4

The water drain discharge totals 12.6 kmol/h of water (226.8 kg/h), which is well within the maximum permitted discharge limit of 550 kg/h.

4. Energy Balances

Energy balances were performed for four process units using the material balance results as a basis. All enthalpies are referenced to the standard state (298.15 K, 1 bar) with species in the gas phase. Cp data and enthalpies of formation from the project data sheet were used throughout. Results are reported in kJ/s (kW).

4.1 Chlorination Reactor Energy Balance

Principal Author: Jibran Ahmed (2449068)

Checked by: Mustafa Qureshi (2543631)

The chlorination reactor (R1+R2 combined) operates isothermally at 270°C (543.15 K). All feeds are determined to enter at the reactor temperature. The energy balance therefore simplifies only to the enthalpy of reaction. Two reactions proceed in parallel: chlorination (Equation 1) and benzene combustion (Equation 2).

Standard enthalpies of reaction at 298.15 K were calculated using Hess's Law from the enthalpies of formation:

Reaction	ΔH° (kJ/kmol)	Rate (kmol/s)
Chlorination: $C_6H_6 + \frac{1}{2}O_2 + HCl \rightarrow C_6H_5Cl + H_2O$	-180,690	0.01022
Combustion: $C_6H_6 + \frac{1}{2}O_2 \rightarrow 6CO_2 + 3H_2O$	-3,171,540	0.000889

Total reaction heat flow = $(-180,690 \times 0.01022) + (-3,171,540 \times 0.000889) = -1,848 + (-2,819) = -4,667$ kJ/s

The overall duty also accounts for the small difference in stream enthalpies entering and leaving the reactor at 543.15 K. Accounting for all species sensible heats, the energy balance gives:

Total chlorination reactor duty (Q) $\approx -4,735$ kJ/s

The negative sign indicates that heat must be removed from the reactor (exothermic overall). A cooling system capable of removing at least 4,735 kW is required.

Due to the high isothermal operating temperatures standard cooling water cannot be safely used. The heat removal will instead be managed using a high-temperature heat transfer fluid or by vaporising the boiler feedwater into high pressure steam.

4.2 Hydrolysis Reactor Energy Balance

Principal Author: Noel Renni (2517447)

Checked by: Mustafa Abdelkarim (2512713)

The hydrolysis reactor (R3) operates isothermally at 310°C (583.15 K) with all feeds being preheated to reactor temperature. The energy balance again reduces to the enthalpy of the hydrolysis reaction (Equation 3):

Equation 3: $C_6H_5Cl + H_2O \rightarrow C_6H_5OH + HCl$

$\Delta H^\circ (298.15 \text{ K}) = \Delta H_m(\text{phenol}, g) + \Delta H_m(\text{HCl}, g) - \Delta H_m(\text{MCB}, g) - \Delta H_m(\text{H}_2\text{O}, g)$

$= (-119,634) + (-92,360) - (51,870) - (-242,000) = -21,864$ kJ/kmol

Correction to 583.15 K using ΔC_p integration gives $\Delta H^\circ (583.15 \text{ K}) \approx -14,595$ kJ/kmol. The MCB conversion rate is 36.36 kmol/h = 0.01010 kmol/s.

Total hydrolysis reactor duty (Q) = $-14,595 \text{ kJ/kmol} \times 0.01010 \text{ kmol/s} \approx -147.41 \text{ kJ/s}$

The reactor is exothermic and requires cooling duty of approximately 147 kW.

Similarly to the chlorination stage the high operating temperature would require the use of a specialised high temperature heat transfer fluid rather than standard cooling water.

4.3 Waste-Heat Boiler Energy Balance

Principal Author: Mustafa Qureshi (2543631)

Checked by: Jibran Ahmed (2449068)

The waste heat boiler cools the combined reactor discharge from 270°C (543.15 K) to 150°C (423.15 K). The heat released by the process gas is used to generate dry saturated steam at 100°C from feed water at 100°C (the water is assumed to enter at its boiling point).

Boiler duty (heat extracted from process side) = **1,287 kJ/s**

Using the latent heat of vaporisation of water at 100°C ($\Delta H_{vap} = 40,683 \text{ kJ/kmol}$):

Steam generation rate = $(1,287 \text{ kJ/s} \times 3600 \text{ s/h}) / 40,683 \text{ kJ/kmol} = 113.9 \text{ kmol/h} = 2,052 \text{ kg/h}$ of dry saturated steam.

This steam is directly routed to HEX-1 to preheat the recycled HCl stream, eliminating the need for external heating.

4.4 HCl Preheater Heat Exchanger (HEX-1) Energy Balance

Principal Author: Mustafa Abdelkarim (2512713)

Checked by: Noel Renni (2517447)

HEX-1 uses the dry saturated steam from the boiler to preheat the recycled HCl stream from 50°C (323.15 K) to 90°C (363.15 K) before it enters the chlorination reactors. The HCl molar flow rate is $36.80 \text{ kmol/h} = 0.01022 \text{ kmol/s}$, and sensible heat calculations using the Cp polynomial for HCl gas give:

gas-phase Cp data was used for HCl throughout this calculation as no liquid-phase heat capacity data was provided for HCl in the project data sheet. Given that HCl exists predominantly as a gas under the process conditions given, this is considered an appropriate approximation.

HEX-1 duty = **17.2 kW (17.2 kJ/s)**

This is a relatively small duty compared to the reactor energy requirements, confirming that the steam generated by the boiler is more than sufficient to supply HEX-1.

4.5 Product Cooler Heat Exchanger (HEX-2) Energy Balance

Principal Author: Mustafa Qureshi (2543631)

Checked by: Noel Renni (2517447)

After the waste-heat boiler, HEX-2 is used for further cooling the combined reactor products from 150°C down to the target temperature of 65°C before the venting stage. Using the component molar flow rates from the reactor discharge and their respective sensible heat capacities, the energy balance yields:

HEX-2 cooling duty = **727 kW (727 kJ/s)**

As the temperature profile of this unit drops the process stream to 65°C, this 727 kW heat removal requirement will be done using standard cooling water.

5. Distillation Column K2 Design – VLE Analysis

5.1 Background and Theory

Principal Author: Jibran Ahmed (2449068), Noel Reni (2517447), Mustafa Qureshi (2543631), Mustafa Abdelkarim (2512713)

Checked by: All team members checked each other's individual sections.

Distillation column K2 separates the binary benzene/monochlorobenzene (MCB) mixture produced at the top of K1 into a benzene-rich distillate (97 mol% benzene) recycled to the chlorination reactors, and an MCB-rich bottoms stream (97 mol% MCB, i.e. 3 mol% benzene) fed to the hydrolysis reactor. Since benzene (NBP = 80.0°C) is considerably more volatile than MCB (NBP = 131.5°C), separation by distillation is well-suited to this system.

The analysis uses the McCabe-Thiele graphical method, which determines the number of theoretical stages required for a given separation. The method relies on the vapour-liquid equilibrium (VLE) relationship and the material balance relationships (operating lines) for the rectifying and stripping sections of the column.

For this ideal binary system, Raoult's Law was applied to generate the VLE data:

$$y^l = x^l \cdot P^{l*}(T) / P^{TOT}$$

where y^l is the vapour mole fraction, x^l is the liquid mole fraction, $P^{l*}(T)$ is the saturation pressure at temperature T (K) from the Antoine equation, and $P^{TOT} = 750.062$ mmHg (1 bar). The Antoine equation used is of the natural-log form:

$$\ln(P^* / \text{mmHg}) = A - B / (T/K + C)$$

Table 5.1: Antoine Constants for Benzene and MCB

Species	A	B	C (K)
Benzene	15.9008	2788.51	-52.36
MCB (Chlorobenzene)	16.0676	3295.12	-55.60

The bubble-point temperature at each liquid composition x was solved numerically from Raoult's Law total pressure condition ($x^l \cdot P^{l*} + x^{MNI} \cdot P^{MNI*} = P$). The equilibrium vapour composition was then calculated. The relative volatility α of benzene over MCB ranges from approximately 4.8 at $x^l = 0$ (pure MCB boiling) to 3.0 at $x^l = 1$ (pure benzene boiling), confirming that the system is amenable to distillation with a moderate number of stages.

5.2 Operating Lines and Minimum Reflux

The feed to K2 is assumed to be a saturated liquid ($q = 1$) at its bubble-point temperature, giving a vertical q -line at $x = z^f$. For $q = 1$, the pinch point occurs at $x = z^f$ on the equilibrium curve, and the minimum reflux ratio is:

$$R_{min} = (x^D - y^*) / (y^* - z^f)$$

where $y^* = y^e_v(z^f)$ is the equilibrium vapour composition at the feed liquid composition.

The rectifying operating line (ROL) connects (x^D, x^D) and has slope $R/(R+1)$. The stripping operating line (SOL) passes through (x^l, x^l) and intersects the ROL at the q -line. Theoretical stages were

stepped off using the McCabe-Thiele procedure: starting from the distillate composition, alternating between the equilibrium curve (horizontal steps) and the appropriate operating line (vertical steps), switching from ROL to SOL at the feed stage when x falls below z^f . The total number of steps, including the reboiler, gives the theoretical stage count.

5.3 K2 Column Design – Team Diagram

Principal Author: Jibran Ahmed (2449068), Mustafa Qureshi (2543631), Noel Renni (2517447), Mustafa Abdelkarim (2512713)

The actual feed composition to K2, determined from the material balance, is $z^f = 0.6084$ mol fraction benzene (39.2 mol% MCB). The operating reflux ratio was set at 1.5 times the minimum, which is a common initial design choice that provides a reasonable balance between capital cost (number of stages) and operating cost (energy for reboiler and condenser). Table 5.2 summarises the key design parameters.

Table 5.2: K2 Column Design Parameters (Team Diagram)

Parameter	Value	Units
Feed composition, z_F (benzene)	0.6084	mol fraction
Feed composition (MCB, $1 - z_F$)	0.3916	mol fraction
Feed condition, q	1.0 (sat. liquid)	-
Distillate composition, x_D	0.97	mol fraction benzene
Bottoms composition, x_B	0.03	mol fraction benzene
System pressure, P	750.062 (1 bar)	mmHg
Minimum reflux ratio, $R_{\min}$	0.326	-
Operating reflux ratio, R	0.489	-
$R / R_{\min}$	1.50	-
Theoretical stages (incl. reboiler)	9	-
Theoretical trays (excl. reboiler)	8	-

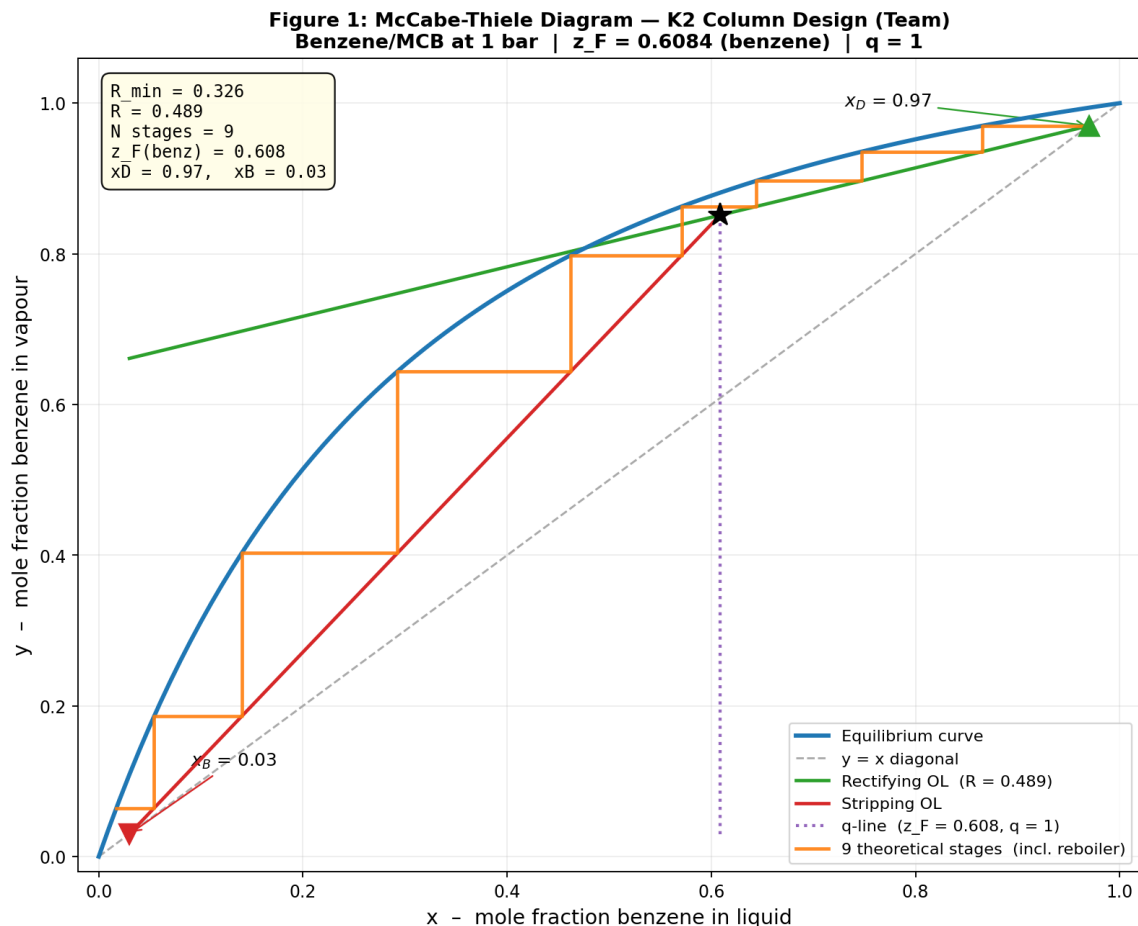


Figure 1: McCabe -Thiele Diagram - K2 Column Design (Team). 9 theoretical stages at $R = 0.489$ ($1.5 \times R_{\min}$). $z_F = 0.6084$, $x_D = 0.97$, $x_B = 0.03$, $q = 1$.

5.4 Individual Analyses – Effect of Feed Composition on Required Reflux Ratio

The purpose of the individual analyses is to assess how column K2 would respond if the per-pass conversion of benzene in R1 were improved. Higher conversion would produce more MCB per pass, increasing the MCB content of the K2 feed. Using the same number of theoretical stages ($N = 9$, established in the team design), each team member determined the reflux ratio required to maintain the product specifications ($x_D = 0.97$ and $x_B = 0.03$) for a different feed composition (20, 30, 40, and 50 mol% MCB). In all scenarios, the feed is at its bubble point ($q = 1$).

5.4.1 Individual VLE: 20 mol% MCB Feed (Jibran Ahmed, 2449068)

At 20 mol% MCB ($z_F = 0.80$ benzene), the feed is benzene-rich, and the column must work harder in the stripping section to recover MCB-lean bottoms. The minimum reflux ratio is low ($R_{\min} = 0.118$) because the feed is already close to the distillate specification. With 9 stages, a reflux ratio of only $R = 0.165$ is required.

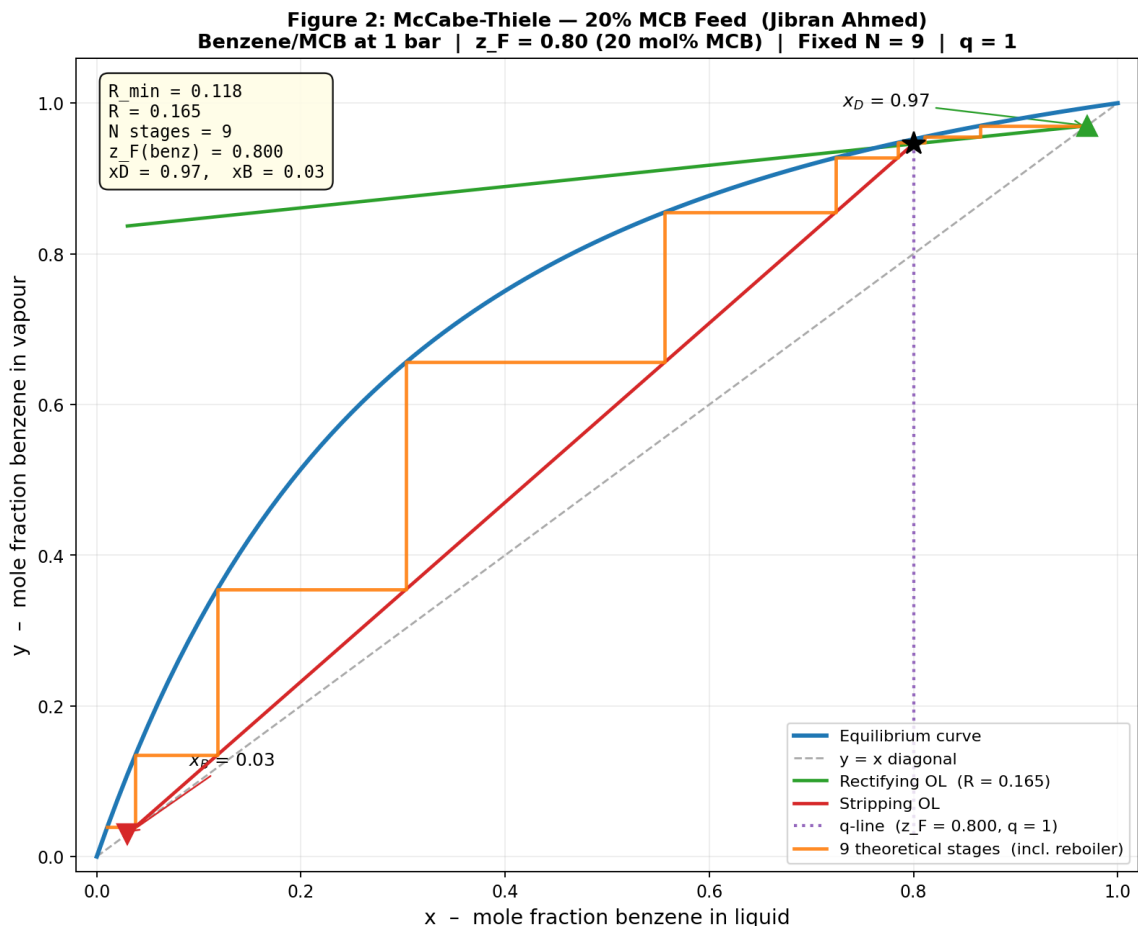


Figure 2: McCabe - Thiele Diagram - 20 mol% MCB Feed (Jibran Ahmed). $R = 0.165$ ($1.40 \times R_{\min}$), $N = 9$ stages. $x_D = 0.97$, $x_B = 0.03$, $z_F = 0.80$.

5.4.2 Individual VLE: 30 mol% MCB Feed (Mustafa Qureshi, 2543631)

At 30 mol% MCB ($z_F = 0.70$ benzene), the required reflux ratio rises to $R = 0.325$. This reflects the increased rectifying-section workload as the feed becomes richer in MCB.

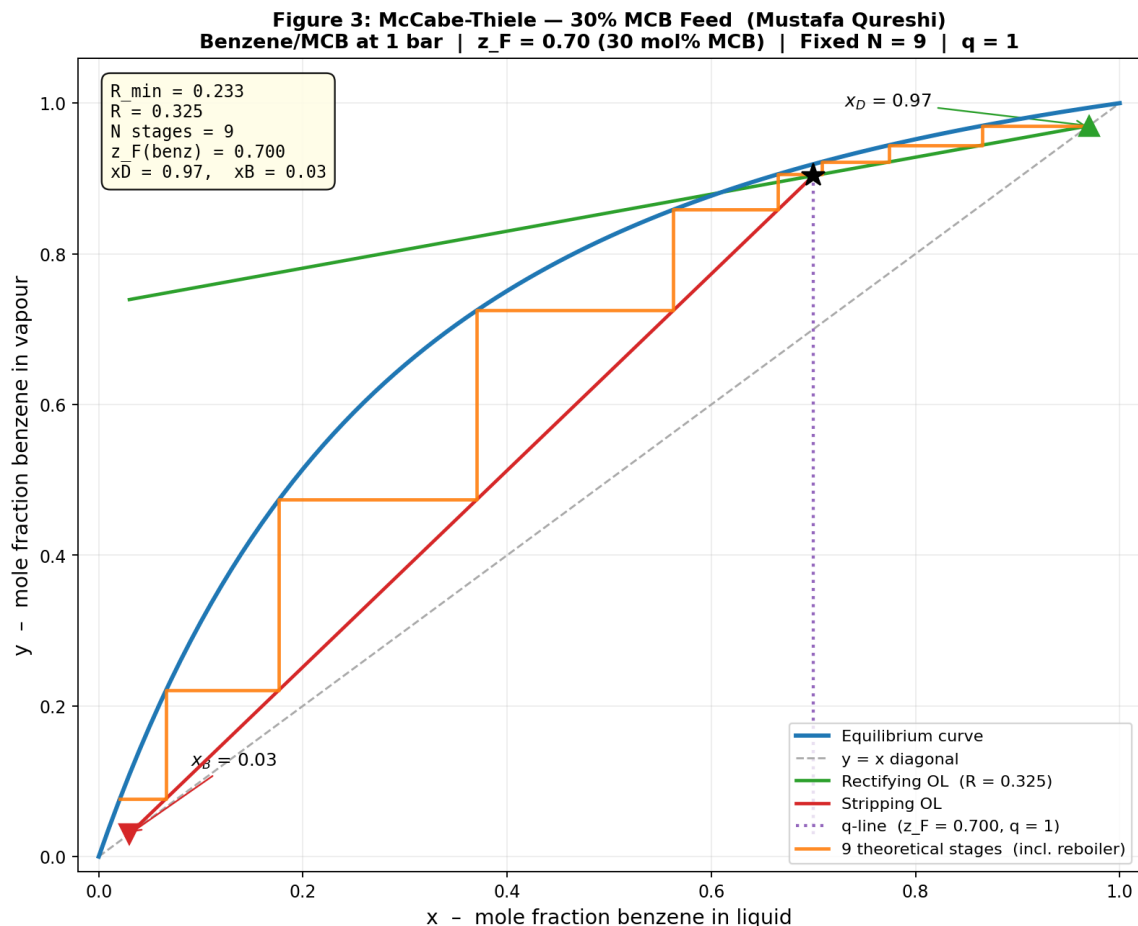


Figure 3: McCabe -Thiele Diagram - 30 mol% MCB Feed (Mustafa Qureshi). $R = 0.325$ ($1.40 \times R_{min}$), $N = 9$ stages. $x_D = 0.97$, $x_B = 0.03$, $z_F = 0.70$.

5.4.3 Individual VLE: 40 mol% MCB Feed (Noel Reni, 2517447)

At 40 mol% MCB ($z_F = 0.60$), which is close to the actual process feed composition, $R = 0.469$ is required. This is very close to the team design value (0.489), which provides a useful cross-check and demonstrates the internal consistency of the analysis.

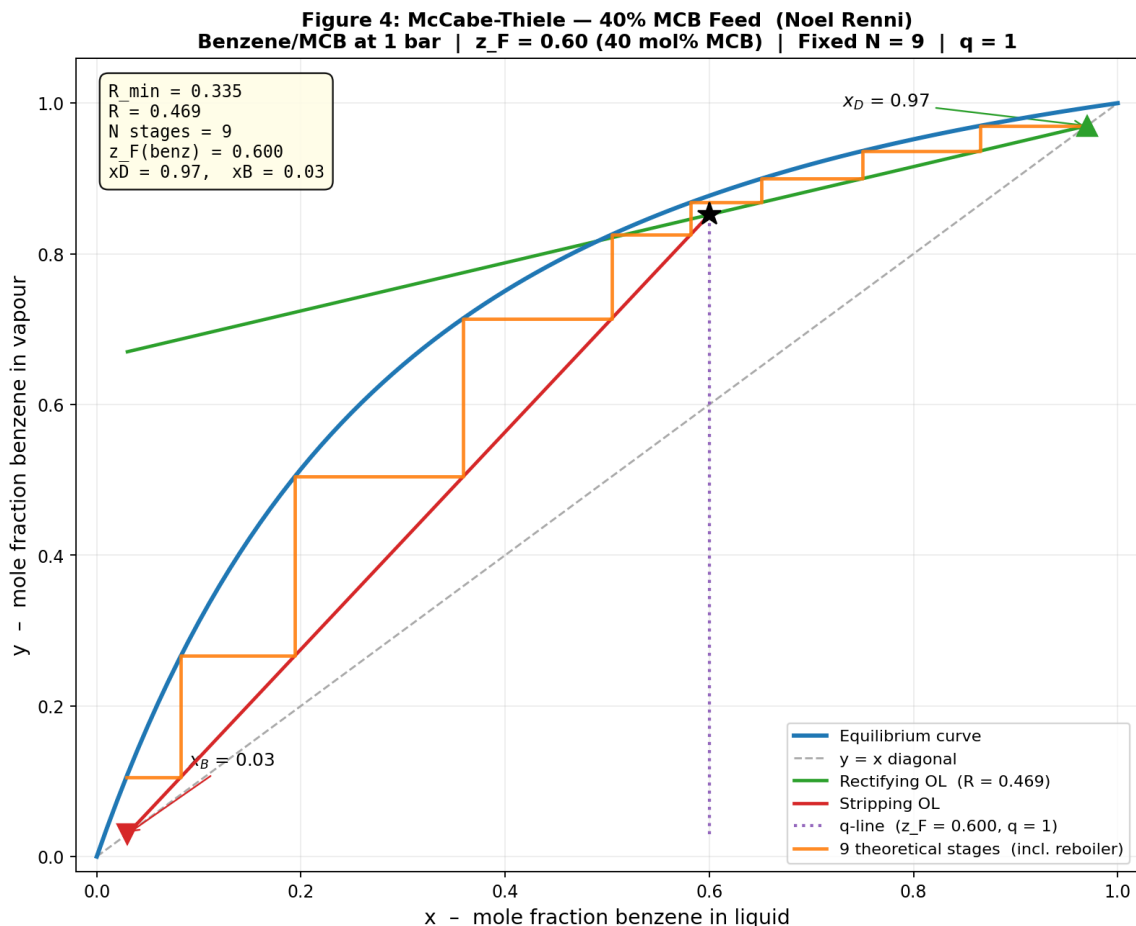


Figure 4: McCabe -Thiele Diagram - 40 mol% MCB Feed (Noel Renni). $R = 0.469$ ($1.40 \times R_{\min}$), $N = 9$ stages. $x_D = 0.97$, $x_B = 0.03$, $z_F = 0.60$.

5.4.4 Individual VLE: 50 mol% MCB Feed (Mustafa Abdelkarim, 2512713)

At 50 mol% MCB ($z_F = 0.50$), the required reflux ratio increases to $R = 0.726$ ($1.59 \times R_{\min}$). The rectifying section now needs to achieve a much larger enrichment, requiring significantly more vapour flow (and hence reflux) to maintain the distillate specification.

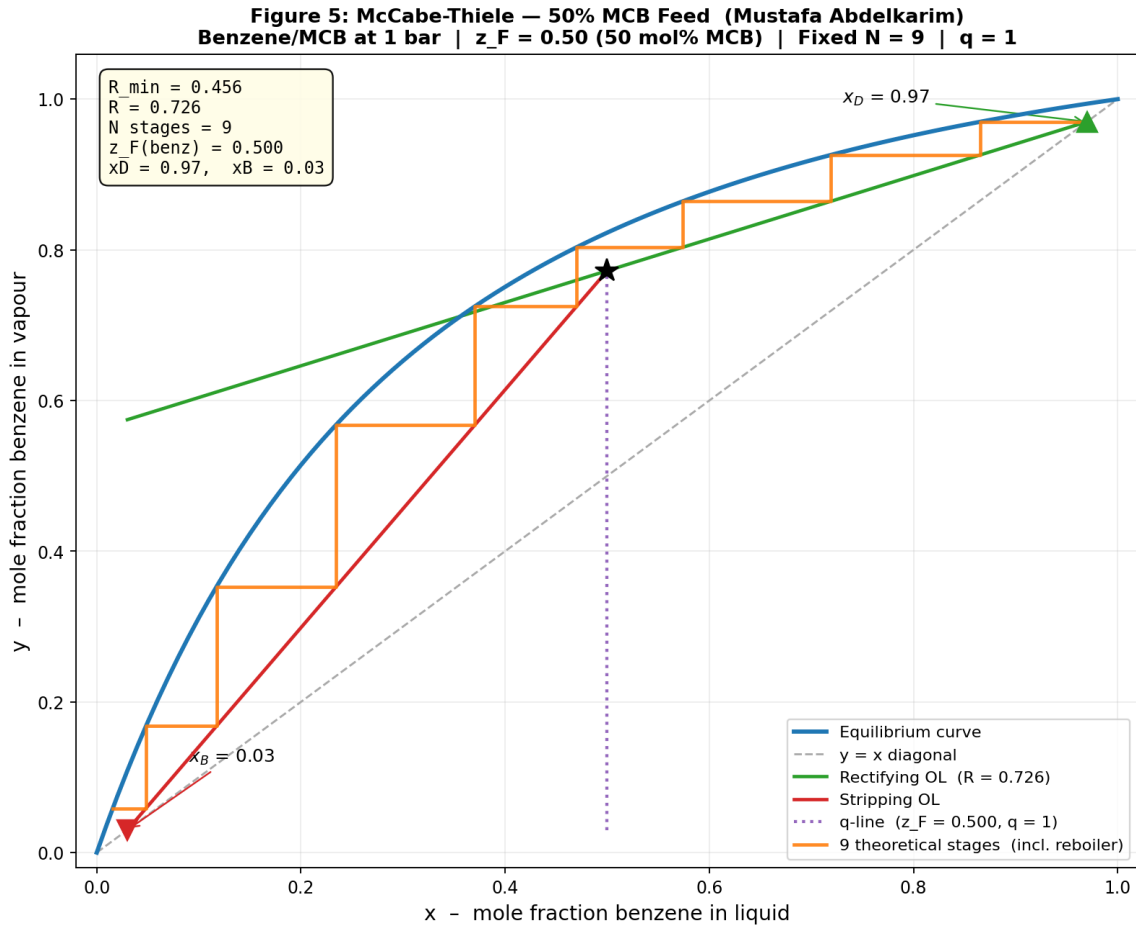


Figure 5: McCabe–Thiele Diagram – 50 mol% MCB Feed (Mustafa Abdelkarim). $R = 0.726$ ($1.59 \times R_{min}$), $N = 9$ stages. $x_D = 0.97, x_B = 0.03, z_F = 0.50$.

5.5 Discussion and Column Design Conclusions

Table 5.3: Summary of VLE Analysis Results for All Scenarios

Scenario	z_F (benzene)	R_{min}	R (operating)	R/R_{min}	N stages
Team Design (actual)	0.6084	0.326	0.489	1.50	9
20% MCB (Jibran)	0.80	0.118	0.165	1.40	9
30% MCB (Mustafa Q)	0.70	0.233	0.325	1.40	9
40% MCB (Noel)	0.60	0.335	0.469	1.40	9
50% MCB (Mustafa A)	0.50	0.456	0.726	1.59	9

The results shows a clear and physically meaningful trend: as the MCB content of the feed increases (representing higher reactor conversion), the required reflux ratio increases monotonically. This is because the rectifying section must perform a larger separation task to produce a 97 mol% benzene distillate from an increasingly MCB-rich feed. At 50 mol% MCB, the required reflux is 0.726, which is 48.5% higher than at the current process conditions (0.489). This has direct implications for energy consumption, as a higher reflux ratio demands greater reboiler and condenser duties.

This confirms that the K2 column, as designed with 9 theoretical stages (8 trays plus reboiler), is capable of meeting the required product specifications across all feed composition scenarios

explored, provided that the reflux ratio is adjusted accordingly. From an operational standpoint, this gives useful flexibility: if the chlorination reactors are run at higher conversion (improving MCB yield), the column can still deliver the required separation with an appropriately increased reflux ratio. The column would therefore not need to be redesigned, only re-operated.

Regarding the optimum reflux ratio for the actual process feed ($z_F = 0.6084$): the operating ratio of $R = 0.489$ ($1.5 \times R_{\min}$) is a reasonable and standard engineering choice. Operating below $R_{\min}$ is impossible (infinite stages would be required), and operating near $R_{\min}$ leads to a very large number of stages. As R increases above $R_{\min}$, the stage count decreases but operating costs increase. The conventional optimum lies in the range 1.2 to 1.5 times $R_{\min}$, and our chosen value of 1.5 provides a sensible margin above minimum reflux with a manageable stage count of 9.

6. Safety and Environmental Analysis

Principal Author: Noel Renni (2517447), Mustafa Abdelkarim (2512713)

6.1 Safety Analysis

The process handles several hazardous chemicals at elevated temperatures and pressures. A robust safety management system is essential to mitigate these hazards. Key hazards are summarised below.

6.1.1 Benzene

Benzene is a Group 1 carcinogen (IARC, 2012) with a UK Workplace Exposure Limit (WEL) of 1 ppm (8-hour TWA). It is highly flammable (flash point -11°C , flammable limits 1.2–79 vol% in air) and its vapour is denser than air, creating potential for accumulation in low-lying areas. All benzene handling areas must be classified as hazardous zones, with explosion-proof electrical equipment. Workers must use respiratory protection and minimise skin contact. Continuous atmospheric monitoring and fixed hydrocarbon detectors should be installed around benzene storage and process areas.

6.1.2 Monochlorobenzene (MCB)

MCB is toxic (harmful if inhaled or ingested) with an irritating vapour. It is also flammable (flash point 29°C , flammable limits 1.3–11 vol% in air). Given that it is used as both an intermediate and a product stream, a significant quantity is present throughout the plant. Safe design requires ventilation, vapour extraction, and leak detection. MCB should be stored in sealed, earthed tanks away from ignition sources.

6.1.3 Hydrochloric Acid (HCl)

Both the liquid 25 wt% HCl feed and gaseous HCl produced in the process are highly corrosive. HCl is severely damaging to the respiratory system, eyes and skin. The process must use materials of construction resistant to HCl (e.g. PTFE-lined pipework, Hastelloy C or glass-lined vessels where appropriate). Automatic safety showers and eyewash stations should be installed near all HCl handling areas. HCl gas detectors with audible and visual alarms should be installed in all areas where HCl may be present. Emergency neutralisation provisions (e.g. scrubbers using caustic soda) should be in place for any vent or relief streams.

6.1.4 Phenol

Phenol is toxic and can be absorbed through the skin, causing severe effects. It has a melting point of 40.8°C , so it may solidify in pipework during shutdowns, creating blockage hazards. Workers handling phenol must wear impermeable gloves and protective clothing. Process pipework carrying phenol should be heat-traced to prevent solidification. Emergency washing facilities must be immediately available.

6.1.5 High-Temperature Operations

The chlorination reactors operate at 270°C and the hydrolysis reactor at 310°C . These temperatures are well above the temperatures of benzene (498°C) and MCB (640°C), but the high temperatures combined with pressurised equipment pose significant burn and explosion risks. All high-temperature pipework and equipment must be insulated and lagged. Regular inspection schedules should verify the integrity of seals, gaskets, and insulation. Safety relief valves and rupture disks must be specified for all pressurised vessels.

6.2 Environmental Analysis

The process generates several waste streams that could cause environmental harm if released uncontrolled.

6.2.1 Water Discharge

The overflow from the water holding vessel discharges to drain at 226.8 kg/h, well within the 550 kg/h limit. However, this water may contain trace quantities of benzene, MCB, HCl, or phenol from the decanter and upstream operations. Before discharge, the water stream should be tested and, if necessary, treated (e.g. by steam stripping, activated carbon adsorption etc) to meet consent conditions and prevent harm to aquatic ecosystems. Phenol in particular is highly toxic to aquatic organisms even at very low concentrations.

6.2.2 Vent Gas Emissions

The vented gases from the separator vessel contain primarily N_2 , O_2 , and CO_2 , with the assumption that no volatile organic compounds (VOCs) leave through the vent. In practice, trace quantities of benzene and other VOCs are likely to be present, requiring a vent condenser or scrubber system to remove them before atmospheric release. CO_2 is a greenhouse gas and its emission should be minimised where possible.

6.2.3 Purge Stream

The hydrolysis recycle purge stream ($r = 0.995$, i.e. 0.5% of unconverted MCB) prevents build-up of volatile impurities. This stream contains MCB and potentially benzene. It must be routed to a suitable treatment unit rather than being vented directly.

6.2.4 Spill and Containment

All storage tanks and process vessels should be surrounded by bunded containment areas designed to hold at least 110% of the largest vessel capacity. Drains within the bunded area should lead to a collection vessel rather than to the main drain network. Emergency response plans should include procedures for spill containment and neutralisation of HCl, and for decontamination of soil and water in the event of a major release.

7. Conclusions and Recommendations

Principal Author: All team members

This report has presented a complete preliminary design of a continuous phenol production process. The key conclusions and recommendations are as follows.

The material balance confirms the feasibility of the process: fresh benzene feed of 41.1 kmol/h (with appropriate HCl and air feeds) achieves approximately 97.2% overall benzene conversion to produce 36.4 kmol/h of phenol, with a small quantity of benzene exiting in the K2 bottoms stream. The water drain discharge (226.8 kg/h) satisfies the environmental constraint of 550 kg/h. The design relies on effective recycling of benzene (from K2 distillate) and HCl (from the hydrolysis reactor), which minimises raw material costs.

The energy balances reveal that both reactors are strongly exothermic and require effective heat removal. The waste-heat boiler recovers 1,287 kJ/s from the chlorination reactor outlet to produce 2,052 kg/h of steam, which is used in HEX-1 to preheat the HCl recycle stream. This is a positive feature of the design, demonstrating good energy renewability.

The VLE analysis confirms that K2 can achieve the required benzene/MCB separation with 9 theoretical stages (8 trays plus reboiler) at $R = 0.489$. The individual analyses demonstrate that if reactor conversion improves and the feed becomes richer in MCB (up to 50 mol%), the column can still achieve the required separation by increasing the reflux ratio to 0.726, without physical modification. This provides valuable operational flexibility for future process optimisation.

Several limitations and recommendations should be noted. The material and energy balances assume ideal behaviour and simplified reactor kinetics (fixed conversion, no by-product formation beyond the specified combustion reaction). In practice, more rigorous modelling and pilot-plant data would be required before scale-up. The use of ideal Raoult's Law in the VLE analysis is a simplifying assumption; for greater accuracy, activity coefficient models should be evaluated. The PFD and BFD presented here are preliminary and should be refined with proper process simulation software. The safety and environmental analysis is introductory; a full HAZOP and environmental impact assessment would be required for detailed design.

8. References

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Appendix A – Meeting Minutes

Meeting 1 – Initial Project Kick-Off

Date & Time	15 February 2026, 14:00
Location/Platform	MS Teams
Present	Mustafa Qureshi, Jibran Ahmed, Noel Renni, Mustafa Abdelkarim
Apologies	None
Minutes	77

Brief Summary of Discussion

- The team reviewed the EGA-102 design brief for phenol production. Discussed the two-step chlorination and hydrolysis route.
- Divided initial research responsibilities. Agreed that mass balances will be the immediate priority once the kinetic data is confirmed.
- Established a team WhatsApp group for daily comms and a shared OneDrive folder for the Excel sheets and Word document.

Action	By Whom
Set up shared OneDrive directory	Jibran Ahmed
Review standard enthalpies of formation for components	Noel Renni
Draft initial block flow sequence (BFD)	Mustafa Qureshi
Extract conversion limits from the design brief	Mustafa Abdelkarim

Meeting 2 – Process Route & Kinetic Data Review

Date & Time	4 March 2026, 15:00
Location/Platform	MS Teams
Present	Mustafa Qureshi, Jibran Ahmed, Noel Renni, Mustafa Abdelkarim
Apologies	None
Minutes	78

Brief Summary of Discussion

- Reviewed the reactor specifications from the brief. Confirmed that the chlorination reactors (R1 and R2) will operate in parallel, each achieving 40% conversion of benzene per pass.

- Discussed how to handle the complex mixture exiting the reactors. Agreed that a decanter is needed to split the aqueous phase from the organic phase before sending the organics to the extraction unit.
- Reviewed the initial BFD sketch. Agreed it needs to be simplified to only show the main chemical routing, saving utility lines for the PFD.

Action	By Whom
Update BFD to reflect parallel R1/R2 setup	Mustafa Qureshi
Begin setting up the Mass Balance spreadsheet framework	Noel Renni & Jibran Ahmed
Compile standard heat of formation data for all species	Mustafa Abdelkarim

Meeting 3 – Mass Balance Review

Date & Time	18 March 2026, 16:00
Location/Platform	Library
Present	Mustafa Qureshi, Jibran Ahmed, Noel Renni, Mustafa Abdelkarim
Apologies	None
Minutes	153

Brief Summary of Discussion

- Reviewed the initial Excel mass balance spreadsheet.
- Identified a complication with the benzene recycle stream coming off the K2 distillation column. Team worked through the algebra to ensure the 100 kmol/h fresh feed basis was maintained while accounting for the 97.2% overall conversion.
- Agreed on the assumption that no volatile organics escape with the vent gases (N_2 and O_2 only) to simplify the separator calculation.

Action	By Whom
Finalise the Excel mass balance formulas	Mustafa Qureshi & Noel Renni
Format Table 3.1 for the report draft	Jibran Ahmed
Begin drafting PFD based on finalised mass streams	Mustafa Abdelkarim

Meeting 4 – PFD Drafting & Distillation Strategy

Date & Time	2 April 2026, 13:00
Location/Platform	Library

Present	Mustafa Qureshi, Jibran Ahmed, Noel Renni, Mustafa Abdelkarim
Apologies	None
Minutes	57

Brief Summary of Discussion

- Team reviewed and officially signed off on the final Excel mass balance results. All streams now close successfully around the 100 kmol/h basis.
- Discussed the sequencing for the distillation columns. Agreed that K1 will drop out the heavy phenol product at the bottom, while K2 will separate the unreacted benzene (top) from the MCB (bottom) for recycling and further processing.
- Transitioned from the BFD to the detailed PFD. Mapped out where heat exchangers will be necessary to condition the feeds between the main unit operations.

Action	By Whom
Draw the first draft of the detailed PFD	Jibran Ahmed
Set up the framework for the Energy Balance Excel sheet	Mustafa Abdelkarim & Noel Renni
Draft the material balance methodology text for the report	Mustafa Qureshi

Meeting 5 – Energy Balance Strategy

Date & Time	12 April 2026, 11:00
Location/Platform	MS Teams
Present	Mustafa Qureshi, Jibran Ahmed, Noel Renni, Mustafa Abdelkarim
Apologies	None
Minutes	104

Brief Summary of Discussion

- Discussed the extreme operating temperatures of R1/R2 (270°C) and R3 (310°C). Recognised that standard cooling water is unviable. Agreed to specify high-temperature heat transfer fluids in the report.
- Explored options for energy recovery. Agreed to implement a Waste-Heat Boiler immediately after the chlorination reactors to generate steam, which will then be used to pre-heat the recycled HCl stream in HEX-1.
- Noted that liquid-phase Cp data for HCl is missing from the data sheet; agreed to use gas-phase data as a stated engineering approximation.

Action	By Whom
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Calculate waste-heat boiler steam generation rate	Mustafa Qureshi
Perform sensible heat calculations for HEX-1	Mustafa Abdelkarim
Draft the reactor energy balance sections	Noel Renni & Jibran Ahmed

Meeting 6 – Final Pre-Submission Review

Date & Time	30 April 2026, 20:00
Location/Platform	Online Voice Call- Discord
Present	Mustafa Qureshi, Jibran Ahmed, Noel Renni, Mustafa Abdelkarim
Apologies	None
Minutes	124
Total Minutes	593

Brief Summary of Discussion

- A phase state discrepancy was spotted in the hydrolysis reactor calculations. Team recalculated this using gas phase constants to ensure accuracy bringing final cooler duty to 147.41 kW.
- Resolved some minor routing errors with the ideal separator to ensure the unconverted MCB routed back correctly to the hydrolysis reactor whilst HCl/phenol routed to extractor.
- Realized raw excel sheets were not being submitted so references to them in the report removed.
- Assigned final completion duties and fixed some formatting in the report (standardizing the numbers in mass balance table to match the PFD).

Action	By Whom
Update hydrolysis Energy Balance maths	Mustafa Qureshi & Mustafa Abdelkarim
Correct BFD routing loop	Noel Renni
Finalise PFD stream numbering and align table 3.1	Jibran Ahmed
Scrub Excel references from report	Mustafa Qureshi
Perform final PDF conversions and submit to canvas	Noel Renni

Appendix B – Team Contributions and Authorship

Table B.1: Section Authorship and Contributions

Report Section	Principal Author(s)	Checked by
Summary	All team members	All team members
Process Overview Description	Jibran Ahmed, Mustafa Qureshi	Noel Renni, Mustafa Abdelkarim
Block Flow Diagram	All team members	All team members
Process Flow Diagram	All team members	All team members
Material Balance Tables	Mustafa Qureshi, Noel Renni	Jibran Ahmed, Mustafa Abdelkarim
Material Balance Explanation	Mustafa Qureshi, Noel Renni	Jibran Ahmed, Mustafa Abdelkarim
Energy Balance – Chlorination Reactor	Jibran Ahmed	Mustafa Qureshi
Energy Balance – Hydrolysis Reactor	Noel Renni	Mustafa Abdelkarim
Energy Balance – Boiler	Mustafa Qureshi	Jibran Ahmed
Energy Balance – HCl Heat Exchanger	Mustafa Abdelkarim	Noel Renni
Column Design: Explanation & Team VLE	All team members	All team members
Individual VLE: 20 mol% MCB	Jibran Ahmed (2449068)	Mustafa Qureshi
Individual VLE: 30 mol% MCB	Mustafa Qureshi (2543631)	Noel Renni
Individual VLE: 40 mol% MCB	Noel Renni (2517447)	Mustafa Abdelkarim
Individual VLE: 50 mol% MCB	Mustafa Abdelkarim (2512713)	Jibran Ahmed
Safety & Environmental Analysis	Noel Renni, Mustafa Abdelkarim	Jibran Ahmed, Mustafa Qureshi
Conclusions and Recommendations	All team members	All team members